JanSaarthi – India's AI Bridge to Governance

Problem Statement

A **significant portion of India's rural and semi-urban population** remains **digitally excluded** due to **limited literacy**, **language diversity**, and **unreliable internet access**. Existing digital services often cater to **English-speaking, tech-savvy users**—**leaving behind those who most need access** to **welfare schemes**, **grievance redressal**, and **legal aid**. This growing **digital divide** demands an **inclusive**, **multilingual**, **multimodal solution**.
JanSaarthi addresses this gap by offering an **AI-powered assistant** accessible via **mobile app**, **WhatsApp**, and **toll-free calls**—delivering **voice**, **text**, and **image-based support** in **regional languages**.
Powered by **Generative AI**, **JanSaarthi ensures every citizen**—regardless of **literacy**, **device**, or **connectivity**—can access **essential government services** in a **human-centric** way.

Target Audience

JanSaarthi is built for **India's rural and underserved citizens** who face challenges like low digital literacy, language diversity, and **limited smartphone or internet access**. These users often struggle with existing government platforms. JanSaarthi empowers them with a simple, multilingual, AI-powered assistant accessible via voice, text, and image—bridging the gap between citizens and public services.

Relevance of Problem

Millions in rural India are excluded from accessing government services due to language barriers, poor digital literacy, and lack of internet. Existing platforms fail to serve these users effectively. **JanSaarthi** addresses this gap with a multilingual, voice-first AI solution—making governance truly inclusive, accessible, and aligned with the vision of Digital India.

Use of Gen-AI

Generative AI forms the backbone of JanSaarthi's inclusive and intelligent architecture. The platform leverages **Bhashini ASR** and Whisper to transcribe regional voice inputs into text, while Bhashini NLP ensures **deep cross-lingual semantic** understanding. **Tesseract OCR** adds multimodal capability by extracting text from scanned documents and images, expanding accessibility beyond typed input. **GPT-4 APIs** handle intent classification and generate context-aware, structured responses. These responses are refined by **fine-tuned Large Language Models** (LLMs) to produce **human-readable** outputs in native languages. Finally, Text-to-Speech (TTS) converts these outputs into natural voice replies, enabling a fully conversational, voice-first experience. This **Gen-AI pipeline** allows JanSaarthi to deliver real-time, personalized civic assistance to every Indian—regardless of literacy, language, or connectivity barriers.

🚀 Solution Framework: How JanSaarthi Works

JanSaarthi runs on a powerful **shared backend system** that supports **three smart user interfaces**:

- 📱 A user-friendly **Mobile App**
- ☎️ A voice-first **Toll-Free IVR system**
- 💬 An accessible **WhatsApp Chatbot**

Together, they ensure **multilingual, seamless access to government services**—no matter the user's literacy level or device.

🎙️ Multiple Ways to Ask, One Intelligent Brain to Understand

Users can interact in the way most natural to them:

- **Speak** in their regional language (processed via **Bhashini ASR** and **Whisper**)
- **Type** text queries
- **Send images or scanned documents** (read using **Tesseract OCR**)

All inputs are passed through **Bhashini NLP**, which performs smart **language detection**, **semantic analysis**, and **normalization**, enabling the system to truly understand regional and non-standard expressions.

🧠 Intelligent Intent Classification

Once a query is processed, it flows through a **GPT-based intent classifier**, which identifies the user's need:

- 🏛️ Searching for **government schemes**
- 🛠️ Filing or tracking a **grievance**
- ⚖️ Seeking **legal assistance**

🔗 Seamless Integration with Government Systems

Based on user intent, JanSaarthi connects to:

- **UMANG/API Setu** for real-time scheme information
- **CPGRAMS** APIs to register and track grievances
- **Firebase Firestore** for legal FAQs and citizen rights resources

🗣️📄💬**Smart, Human-Centered Responses**

A **fine-tuned GPT-4 model** generates clear, contextual, and easy-to-understand responses. These are delivered in formats tailored to the user:

- 🔊 **Voice replies** (via **Bhashini TTS**) for IVR users
- 💬 **WhatsApp or SMS** messages
- 📎 **PDF documents** for official records

💡**Inclusive Features Across All Platforms**

- 📱 **Mobile App**: Offers voice/text/image inputs, user dashboard, and grievance tracking
- ☎️ **Toll-Free IVR**: Enables low-literacy users to talk directly to an AI voice agent using **Asterisk**
- 💬 **WhatsApp Chatbot**: Mirrors app capabilities and connects rural users easily

🔄**Built to Learn & Scale**

JanSaarthi is always evolving. With continuous feedback loops and **AI-driven self-improvement**, the system:

- Learns from real-world usage
- Ensures **real-time synchronization** across platforms
- Delivers consistent and reliable performance—even in **low-connectivity rural areas**

Feasibility

JanSaarthi is built for national-scale deployment with a modular, cloud-native architecture. It seamlessly integrates government-backed platforms like **Bhashini (multilingual ASR/NLP)**, **API Setu (UMANG)**, and **CPGRAMS**, ensuring **regulatory compliance** and smooth interoperability. A **shared backend** powers mobile, WhatsApp, and IVR interfaces uniformly, minimizing engineering effort. Optimized for **low-bandwidth rural areas** via **BharatNet** and **CSC networks**, JanSaarthi is both **inclusive** and **cost-effective**.

Conclusion / MLP & Business Potential

JanSaarthi isn't just a chatbot—it's a civic-tech movement.
It bridges governance and citizens using voice-first, Gen-AI-powered multilingual support. From legal rights to last-mile welfare schemes, JanSaarthi is built as a Minimum Lovable Product (MLP) with a clear path to scale. As a SaaS platform for **government agencies**, JanSaarthi can **license APIs**, power IVR support, and offer regional dashboards for administrators—making it a viable **public-private innovation model for Digital India**.